Technical Note: Inter-Laboratory Differences in Soil Test Data and Implications for Application Error

Space to Place (S2P) Project — Malawi

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1 The Problem

In the S2P trial, soil samples for the three treatment arms were analyzed by two different laboratories:

- **T1 and T2** ($n = 1{,}348$): analyzed by **Cropnuts** (Nairobi)
- **Control** ($n = 416$): analyzed by **FES** (Fertilizer and Environmental Solutions)

Because lab identity is perfectly collinear with treatment status, any systematic difference in how the two labs measure soil nutrients is indistinguishable from a treatment effect when comparing across arms.

This is *not* a concern for the main ITT outcomes (fertilizer quantities, nutrient content, maize yield), which are measured directly in the endline survey and do not depend on soil test accuracy.

However, it *is* a concern for the **application error** outcome. Application error measures the alignment between a farmer's actual fertilizer practices and their soil's measured nutrient deficiencies. Since the "deficiency" side of this measure comes from the soil test, inter-lab bias feeds directly into the outcome variable—and does so differentially by treatment arm.

1.1 Why this matters

Suppose Cropnuts systematically reports higher nutrient levels than FES for the same soil. Then:

- T1/T2 farmers appear to have **smaller deficiencies** (soil already has enough)
- Control farmers appear to have **larger deficiencies** (soil needs more)

If a T1/T2 farmer and a control farmer apply the same amount of fertilizer to identical soil, the T1/T2 farmer's application error will look *smaller*—not because the treatment changed their behavior, but because Cropnuts measured a smaller gap to fill. We would be attributing a lab artifact to the intervention.

Critically, this confound **cannot be resolved by adding controls to the regression**, since lab identity is a perfect predictor of treatment. Any attempt to "control for lab" absorbs the treatment effect entirely.

2 Empirical Evidence of Inter-Lab Differences

The reconciliation report documents the magnitude of these differences across all twelve harmonized soil chemistry variables. To illustrate the range of severity, Figures 1–3 show density

plots for three soil components that span the spectrum from agreement to fundamental incompatibility.

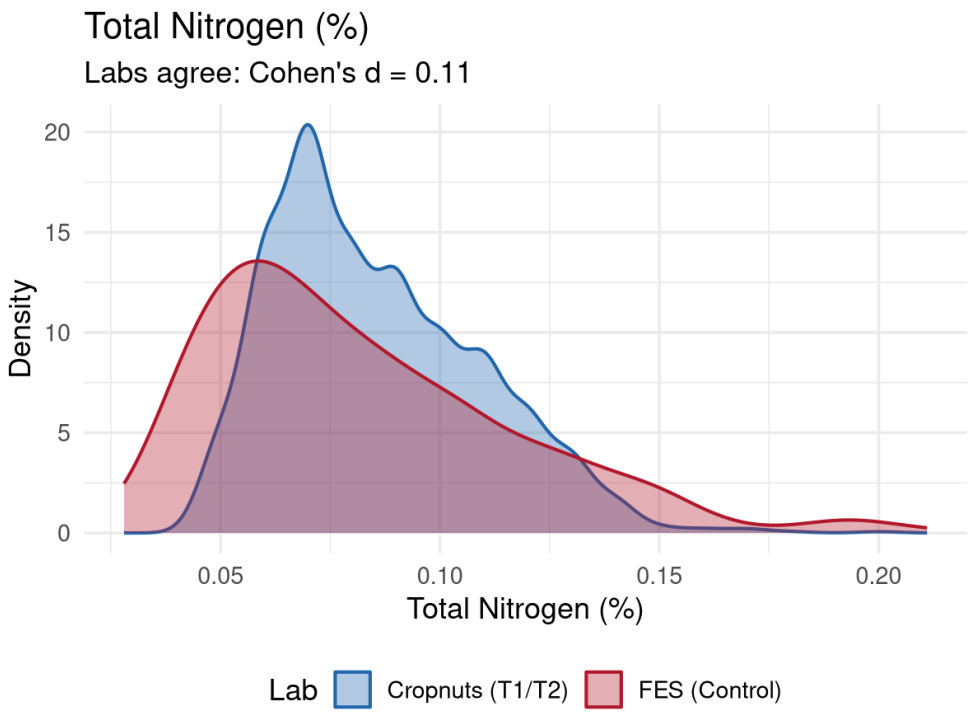

Figure 1: Total Nitrogen (%): distributions from both labs largely overlap. The mean difference is small (Cohen's $d = 0.11$) and not statistically significant at conventional levels. This component is safe for cross-arm comparisons.

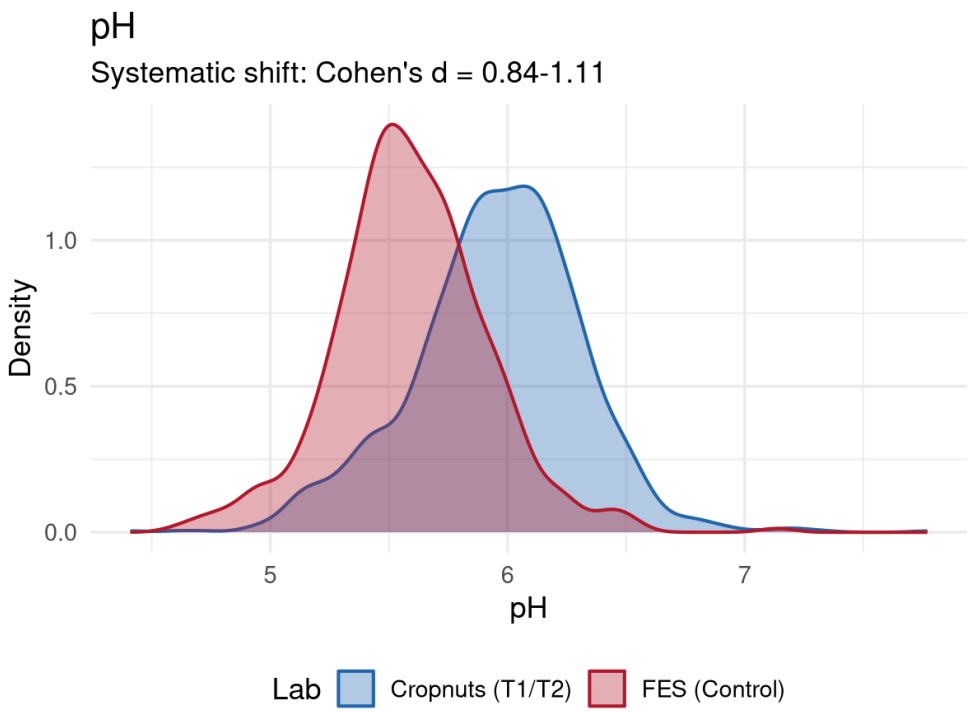

Figure 2: pH: the two distributions have nearly identical shapes but are shifted by approximately 0.3–0.4 units (Cohen's $d = 0.84$–1.11). This is consistent with a calibration offset. Crucially, the rank-ordering of samples is likely preserved.

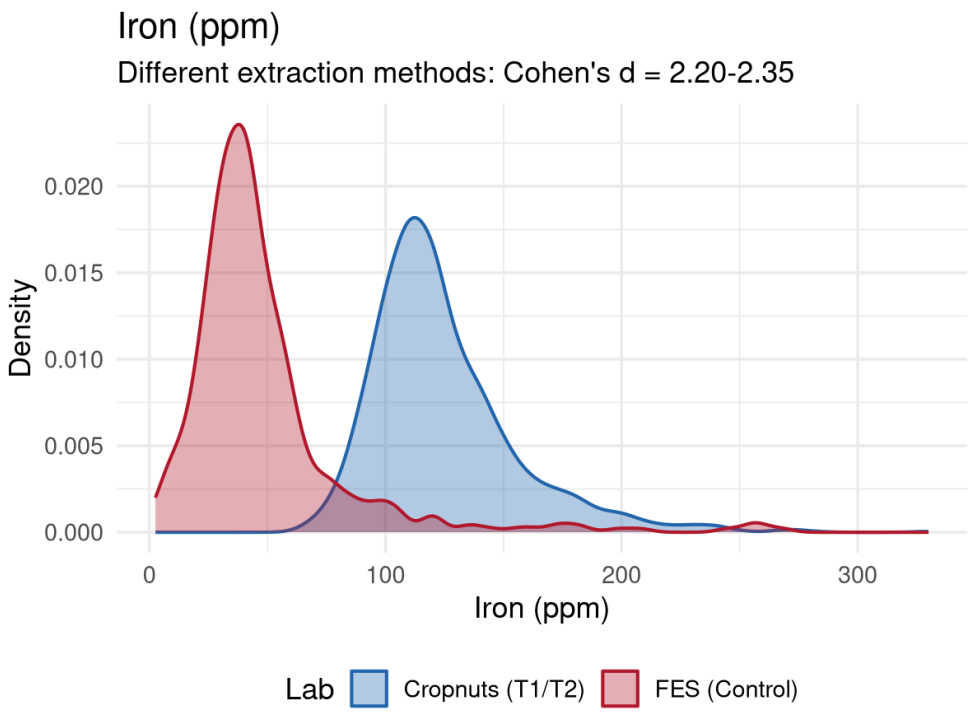

Figure 3: Iron (ppm): distributions are far apart but retain similar right-skewed shapes. Cropnuts reports values roughly 2.5× higher than FES (Cohen's $d = 2.20$–2.35). The visual pattern is consistent with a large multiplicative shift; the statistical distribution alone does not tell us whether the rank-ordering is preserved. That judgement requires knowledge of the underlying extraction methods (see Section 4.5).

2.1 Summary of inter-lab differences

Table 1 classifies all twelve harmonized soil variables by the severity of the lab effect.

Table 1: Classification of soil variables by inter-lab effect size (Cohen's d from T1/T2 vs. Control comparisons).

Category	Variable	Cohen's d
3*Comparable ($d < 0.2$)	Total Nitrogen (%)	0.11–0.15
	Zinc (ppm)	0.02–0.18
	Potassium (ppm)	0.04–0.07
5*Moderate shift ($0.2 < d < 0.7$)	Phosphorus (ppm)	0.17–0.21
	Magnesium (ppm)	0.30–0.35
	CEC (meq/100g)	0.33–0.40
	Copper (ppm)	0.13–0.18
	Boron (ppm)	0.16
4*Large shift ($d > 0.6$)	pH	0.84–1.11
	Calcium (ppm)	0.61–0.66
	Iron (ppm)	2.20–2.35
	Manganese (ppm)	2.38–2.42

3 Proposed Solution: the Monotonicity Assumption

3.1 The assumption

We assume that the inter-lab differences represent a **monotonic transformation** of the true underlying soil values. That is, while the two labs may disagree on the *level* of a nutrient (e.g., Cropnuts reports pH 6.0 where FES would report 5.6), they agree on the *ordering*: if plot A has higher true nitrogen than plot B, both labs will rank A above B.

Formally, let s_i be the true soil nutrient level for farmer i, and let m_i^C and m_i^F be the measurements from Cropnuts and FES respectively. The monotonicity assumption states:

$$s_i > s_j \implies m_i^k > m_j^k \quad \text{for } k \in \{C, F\}$$

This is plausible when the labs use the same or similar extraction methods but differ in calibration, reagent concentrations, or instrument sensitivity. The pH density plot (Figure 2) is a textbook case: the distributions have the same shape, just shifted.

3.2 Why this helps

Under monotonicity, we can replace the *level* of each soil measurement with its **within-lab percentile rank**. Since ranks are invariant to monotonic transformations, the lab calibration effect drops out entirely.

Concretely, for each soil nutrient:

1. Rank all Cropnuts (T1/T2) measurements from lowest to highest and convert to percentiles.
2. Rank all FES (Control) measurements from lowest to highest and convert to percentiles.
3. A farmer at the 80th percentile of nitrogen deficiency in Cropnuts and a farmer at the 80th percentile in FES occupy the same *relative* position in their respective soil quality distribution.

3.3 Constructing the application error measure

Application error can then be defined as the discrepancy between a farmer's **deficiency rank** (from the soil test) and their **application rank** (from the endline survey):

$$\text{Application Error}_i = \left| R_i^{\text{deficiency}} - R_i^{\text{application}} \right|$$

where $R_i^{\text{deficiency}}$ is farmer i's within-lab percentile rank of the measured nutrient deficiency, and $R_i^{\text{application}}$ is their rank of actual fertilizer application (ranked across the full sample, since application is measured identically for all arms).

A farmer who is at the 90th percentile of nitrogen deficiency and the 90th percentile of nitrogen application has an application error near zero—they applied the most because their soil needed the most. This holds regardless of which lab tested their soil.

3.4 What we give up

The rank-based measure is **ordinal, not cardinal**. We can no longer say "this farmer applied 3 kg/acre too little nitrogen." Instead, we say "this farmer is in the 80th percentile of deficiency but only the 40th percentile of application." We lose the physical-units interpretation but gain clean identification across all three treatment arms.

3.5 Which nutrients can use this approach

Selecting which soil variables to include in the rank-based application error analysis requires two distinct criteria:

Criterion 1: Statistical distribution. The density plots allow us to assess whether the two labs' distributions have **similar shapes**. If the shapes are preserved and only the location differs (as in Figures 2 and 3), the visual evidence is consistent with a monotonic shift. Note, however, that even a large magnitude of shift (as with Iron, $d > 2$) does not by itself rule out monotonicity—visually, all three plots in Section 2 show shape-preserved distributions at increasing levels of displacement. The statistical distribution alone cannot confirm or reject rank-preservation; it can only flag variables where the distributions have *different shapes* or where detection limits create floor/ceiling effects that would mechanically break rank-ordering (e.g., many Boron values reported as "< 0.3" in Cropnuts data).

Criterion 2: Domain knowledge about laboratory methods. The more substantive judgement concerns the *extraction and measurement methods* used by each lab. Two labs using the same extraction protocol but different calibration standards would produce a monotonic shift almost by construction. In contrast, labs using fundamentally different extraction chemistry—for example, DTPA versus Mehlich-3 for micronutrients—extract different fractions of the total soil nutrient pool. While both methods may correlate with plant-available nutrients, there is no guarantee that they rank individual soil samples identically. The soil science literature suggests this concern is most acute for Iron and Manganese, where DTPA and Mehlich-3 are known to produce not just level differences but potentially different relative rankings depending on soil pH and organic matter content.

Combined assessment. Applying both criteria:

- **Include with confidence**: Total Nitrogen, Potassium, Zinc—small lab effects ($d < 0.2$), standard methods likely comparable. The rank-based approach works, but is also largely unnecessary since the raw levels are already comparable.
- **Include under monotonicity assumption**: pH, Calcium, Magnesium, CEC—large distributional shifts, but shape-preserved densities and extraction methods that plausibly produce monotonic transformations (same analyte, different calibration).
- **Include with caveats**: Phosphorus—moderate shift, but T1/T2 values are midpoints of ranges rather than exact values, which reduces ranking precision. Copper—small shift but limited variation.
- **Exclude or treat with caution**: Iron, Manganese—although the density plots show shape-preserved distributions (consistent with monotonicity), the likely use of different extraction methods (DTPA vs. Mehlich-3) introduces a domain-knowledge concern that ranking may not be preserved across soil types. Boron—detection-limit floor effects in Cropnuts data mechanically compress ranks at the low end.

4 Alternative Approach: Linear Rescaling

An alternative to the rank-based approach is to **rescale the FES (Control) data to match the Cropnuts (T1/T2) distribution** in mean and standard deviation, and then proceed as if all measurements come from a single lab. For each nutrient, the adjusted FES value would be:

$$\tilde{m}_i^F = \frac{m_i^F - \bar{m}^F}{\sigma^F} \cdot \sigma^C + \bar{m}^C$$

where $\bar{m}^F, \sigma^F$ are the FES sample mean and standard deviation, and $\bar{m}^C, \sigma^C$ are the corresponding Cropnuts statistics. This maps every FES observation onto the Cropnuts scale, preserving each farmer's position relative to their lab's distribution.

4.1 Comparison of the two approaches

The two approaches differ in the strength of their assumptions and what they preserve:

	Rank-based	Linear rescaling
Core assumption	Monotonicity (labs preserve ordering)	Linearity (labs differ by shift + scale)
Interpretation	Ordinal (percentiles)	Cardinal (original units)
Sensitive to outliers	No	Yes
Requires distributional overlap	No	No

Linear rescaling is a *stronger* assumption than monotonicity: it requires that the mapping between labs is not just order-preserving, but specifically of the form $m^C = a + b \cdot m^F$ for constants a and b. Under this assumption, the rescaled data retains its cardinal meaning—application error can be expressed in physical units (e.g., kg/acre of nutrient mismatch), which is more interpretable than percentile gaps.

The linearity assumption is plausible when the lab differences stem from calibration offsets and sensitivity differences (proportional scaling). It is less plausible if the relationship between the two labs' measurements is nonlinear—for instance, if one lab compresses values at the extremes of the distribution. The pH density plot (Figure 2) is reassuring: the two distributions have not only the same shape but also similar spread, suggesting that a simple shift (with $b \approx 1$) may suffice.

4.2 Practical considerations

A practical advantage of linear rescaling is that it can be applied nutrient-by-nutrient using only the sample moments already reported in the reconciliation tables—no additional data are needed. The rank-based approach is similarly straightforward to implement. Both approaches require the same variable-selection step described in Section 4.5 (combining statistical and domain-knowledge criteria), since neither method can fix variables where the labs are measuring fundamentally different analytes.

The two approaches could also be used together: linear rescaling as the primary specification (preserving the cardinal interpretation) and the rank-based measure as a robustness check that relaxes the linearity assumption to monotonicity only.

Note that the T1 vs. T2 comparison on application error remains unaffected by all of this, since both arms were tested by the same lab.

5 Two Distinct Analyses of Application Error

The discussion so far has treated application error as a single concept, but in practice it can be operationalized in two different ways, each answering a different question:

A. Observed behavior vs. soil characteristics. Does the farmer's actual fertilizer application match what their soil *objectively needs*? Here the benchmark is the raw soil

measurement (e.g., soil nitrogen level in ppm or %), and the question is whether farmers who have more deficient soils also apply more fertilizer. The rank-based and linear rescaling approaches described in Sections 3–5 address this analysis.

B. Observed behavior vs. nutrient requirements. Does the farmer's actual application (in kg/ha) match the *recommended requirement* for their soil (also in kg/ha)? This is a more direct measure of application error, since both sides are in the same units and the requirement already accounts for crop type, yield target, and the agronomic translation from soil status to fertilizer need.

Analysis B is arguably more policy-relevant: it captures whether farmers apply the right *amount*, not merely whether they respond to relative soil differences. However, it introduces an additional complication.

5.1 A second layer of lab-specific bias in nutrient requirements

Both Cropnuts and FES report not only raw soil measurements but also derived nutrient requirements (or deficiencies) in kg/ha. A natural approach would be to use these lab-provided requirements directly as the benchmark for Analysis B. However, comparing the inter-lab differences in requirements against those in raw measurements reveals that the labs' agronomic models **amplify rather than absorb** measurement bias for some nutrients:

Table 2: Inter-lab differences (Cohen's d) for raw soil measurements versus lab-computed nutrient requirements. Positive d indicates higher values from Cropnuts.

Nutrient	Raw measurement d	Requirement d	Direction
Nitrogen	+0.15	−0.16	Similar magnitude, reversed sign
Phosphorus	+0.18	+0.56	Amplified
Potassium	+0.05	+0.77	Strongly amplified
Sulfur	—	−0.55	No raw comparison

The potassium case is the most striking: the raw K measurements are nearly identical across labs ($d = 0.05$), yet Cropnuts recommends an average of 52 kg/ha of K_2O while FES reports a deficiency of only 14 kg/ha. The two labs evidently use different agronomic conversion models—different response curves, yield targets, or safety margins—that transform a negligible measurement difference into a large requirement gap.

Using the lab-provided requirements directly for Analysis B would therefore introduce a **second layer of bias**: the first from measurement methodology (Section 2), the second from the agronomic model. For potassium and phosphorus, this second layer dominates.

5.2 Proposed solution: recompute requirements with a single model

To eliminate the model-specific bias, we propose **applying a single, consistent agronomic model to all soil observations**, regardless of which lab produced them. The procedure would be:

1. **Harmonize the raw soil measurements** across labs using the rank-based or linear rescaling approach (Sections 3–5), so that all measurements are on a common scale.
2. **Apply one agronomic conversion model** uniformly to translate the harmonized soil measurements into nutrient requirements (kg/ha). This model should account for crop type and target yield.

3. **Compute application error** as the difference between the uniformly derived requirement and the farmer's actual application from the endline survey.

The agronomic model in step 2 could be derived in several ways:

- **Empirical estimation from Cropnuts data.** We observe both the raw soil inputs and the Cropnuts requirement outputs for over 1,300 T1/T2 farmers. Regressing the requirements on the raw soil variables recovers the Cropnuts conversion model empirically. This fitted model can then be applied to the (rescaled) FES control data to produce comparable requirements.
- **Standard agronomic guidelines.** Published fertilizer recommendation tables for Malawi maize (e.g., from the Guide to Agricultural Production) provide transparent, citable conversion factors that are independent of either lab.
- **Hybrid approach.** Use the standard guidelines as the base model, validated against the Cropnuts outputs to check calibration.

The key advantage of any of these options is that the *same* conversion function is applied to all farmers. This eliminates the second layer of bias entirely: any remaining cross-arm differences in application error reflect genuine differences in farmer behavior, not differences in how the labs translate soil status into recommendations.

5.3 Summary

The two analyses can proceed in parallel:

	Analysis A Behavior vs. soil characteristics	**Analysis B** Behavior vs. requirements
Benchmark	Raw soil nutrient levels	Nutrient requirements (kg/ha)
Lab bias addressed by	Rank-based or linear rescaling	Recomputed with single model
Units of application error	Percentiles or rescaled ppm	kg/ha
Interpretation	Relative alignment	Absolute alignment

Analysis A asks whether farmers *respond to relative soil differences*; Analysis B asks whether they apply the *right amount*. Both are informative, and both require correcting for inter-lab bias—but at different stages of the data pipeline.